

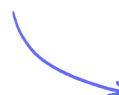
Calculator Questions

Forming & Solving Equations

Forming Equations from Words / Forming Equations from Shapes / Problem Solving with Equations

Medium (10 questions)	/39
Hard (13 questions)	/78
Very Hard (10 questions)	/53
Total Marks	/170

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Medium Questions

- 1 Rangan buys 3.6kg of potatoes and 2.8kg of leeks.

The total cost is \$13.72.

Leeks cost \$2.65 per kilogram.

Find the cost of 1 kg of potatoes.

\$

(3 marks)

- 2 (a) Ahmed sells different types of cake in his shop.

The cost of each cake depends on its type and its size.

Every small cake costs \$ x and every large cake costs $\$(2x + 1)$ The total cost of 3 small lemon cakes and 2 large lemon cakes is \$12.36.

Find the cost of a small lemon cake.

\$

(3 marks)

- (b) The cost of 18 small chocolate cakes is the same as the cost of 7 large chocolate cakes.

Find the cost of a small chocolate cake.

\$ [3]

(3 marks)

- 3 The angles of a quadrilateral are x° , $(x + 5)^\circ$, $(2x - 25)^\circ$ and $(x + 10)^\circ$.

Find the value of x .

$x = \dots\dots\dots$

(3 marks)

- 4 Oranges cost 21 cents each.
Alex buys x oranges and Bobbie buys $(x + 2)$ oranges.
The total cost of these oranges is \$4.20.
Find the value of x .

$x = \dots\dots\dots$

(3 marks)

- 5 (a) Esme buys x magazines at \$2.45 each and y cards at \$3.15 each.

Write down an expression, in terms of x and y , for the total cost, in dollars, of the magazines and the cards.

\$ $\dots\dots\dots$

(2 marks)

- (b) Esme spends \$60.55 in total.
She buys 8 magazines.

How many cards does she buy?

(2 marks)

- 6 i) Factorise $y^2 + 5y - 84$.

[2]

ii)



NOT TO
SCALE

The area of the rectangle is 84 cm^2 .

Find the perimeter.

..... cm [3]

(5 marks)

7 The diagram shows a trapezium.

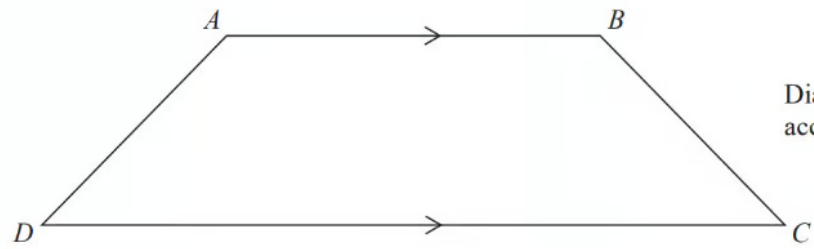


Diagram **NOT**
accurately drawn

$AD = x$ cm.

BC is the same length as AD .

AB is twice the length of AD .

DC is 4 cm longer than AB .

The perimeter of the trapezium is 38 cm.

Work out the length of AD .

(4 marks)

8

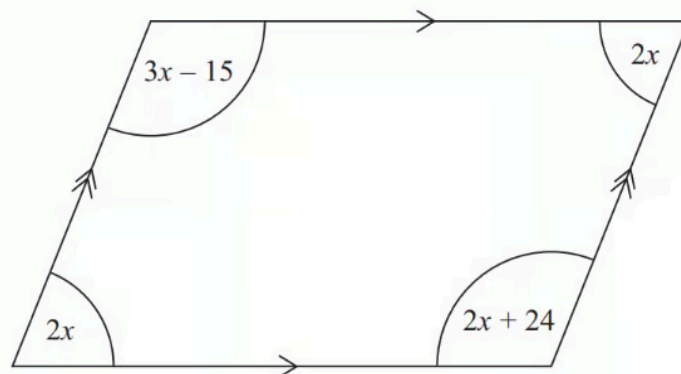


Diagram **NOT**
accurately drawn

The diagram shows a parallelogram.

The sizes of the angles, in degrees, are

$$2x$$

$$3x - 15$$

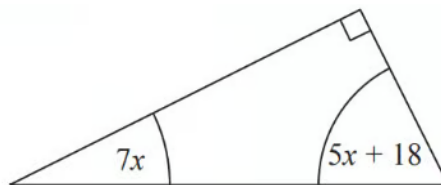
$$2x$$

$$2x + 24$$

Work out the value of x .

(3 marks)

- 9 The diagram shows a right-angled triangle.



All the angles are in degrees.

Work out the size of the smallest angle of the triangle.

(3 marks)

- 10 Redlands School sent x students to a revision day.

St Samuel's School sent twice as many students as Redlands School.

Francis Long School sent 7 fewer students than Redlands School.

Each student paid £ 15 for the revision day.

The students paid a total of £1155

Work out how many students were sent by each school to the revision day.

You must show all your working.

(5 marks)

Hard Questions

- 1 (a)** Paulo and Jim each buy sacks of rice but from different shops. Paulo pays \$72 for sacks costing \$ m each.

Jim pays \$72 for sacks costing \$ $(m + 0.9)$ each.

- i) Find an expression, in terms of m , for the number of sacks Paulo buys.

[1]

- ii) Find an expression, in terms of m , for the number of sacks Jim buys.

[1]

(2 marks)

- (b)** Paulo buys 4 more sacks than Jim.

Write down an equation, in terms of m , and show that it simplifies to

$$10m^2 + 9m - 162 = 0.$$

(4 marks)

- (c)** i) Solve $10m^2 + 9m - 162 = 0$.

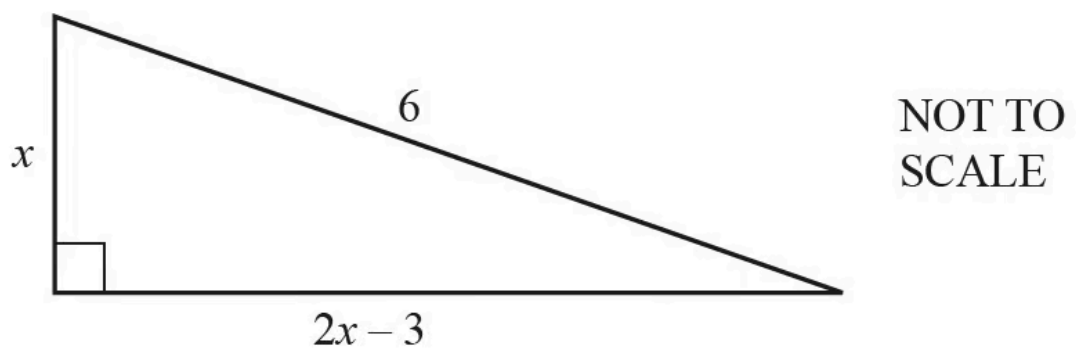
$$m = \dots\dots\dots \text{ or } m = \dots\dots\dots [2]$$

- ii) Find the number of sacks of rice that Paulo buys.

[1]

(3 marks)

2 (a) In this question, all measurements are in metres.



The diagram shows a right-angled triangle.

Show that $5x^2 - 12x - 27 = 0$.

(3 marks)

(b) Solve $5x^2 - 12x - 27 = 0$.

Give your answers correct to 2 decimal places.

$x = \dots\dots\dots$ or $x = \dots\dots\dots$

(2 marks)

(c) Calculate the perimeter of the triangle.

$\dots\dots\dots$ m

(2 marks)

(d) Calculate the smallest angle of the triangle.

(2 marks)

3 (a) The diagram shows a trapezium.

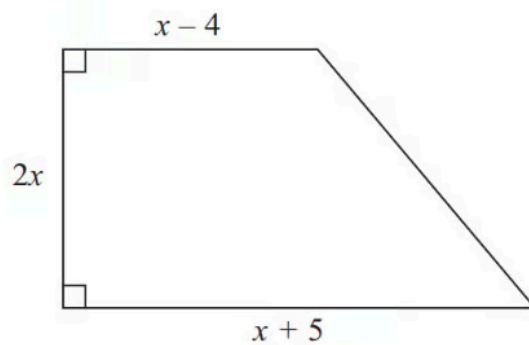


Diagram **NOT**
accurately drawn

All the measurements are in centimetres.

The area of the trapezium is 351 cm^2 .

Show that $2x^2 + x - 351 = 0$

(2 marks)

(b) Work out the value of x .

(2 marks)

4 Solve.

$$\frac{3y}{2y-1} = \frac{3}{4}$$

$y = \dots\dots\dots$

(3 marks)



NOT TO
SCALE

The perimeter of the rectangle is 80 cm.

The area of the rectangle is $A \text{ cm}^2$.

i) Show that $x^2 - 40x + A = 0$.

[3]

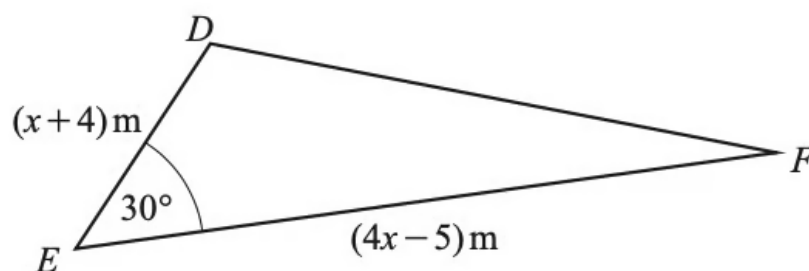
ii) When $A = 300$, solve the equation $x^2 - 40x + A = 0$ by factorising.

$x = \dots\dots\dots$ or $x = \dots\dots\dots$ [3]

iii) When $A = 200$, solve the equation $x^2 - 40x + A = 0$ using the quadratic formula. Show all your working and give your answers correct to 2 decimal places.

$x = \dots\dots\dots$ or $x = \dots\dots\dots$ [4]

6

NOT TO
SCALE

The area of triangle DEF is 70 m^2 .

i) Show that $4x^2 + 11x - 300 = 0$.

[4]

ii) Use the quadratic formula to solve $4x^2 + 11x - 300 = 0$.

Show all your working and give your answers correct to 2 decimal places.

$x = \dots\dots\dots$ or $x = \dots\dots\dots$ [4]

iii) Find the length of DE .

$DE = \dots\dots\dots\text{ m}$ [1]

(9 marks)

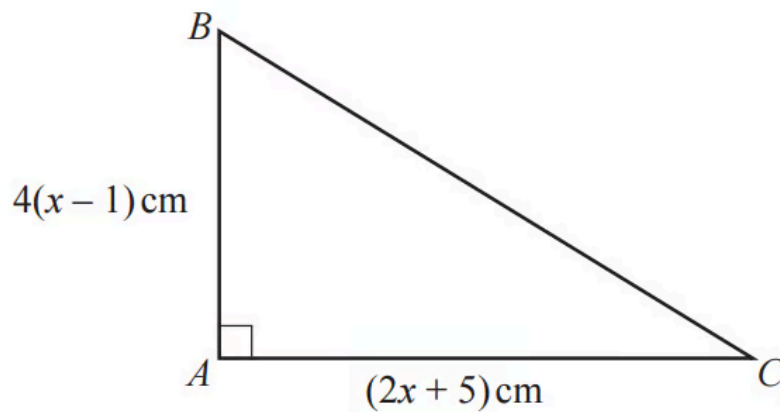
- 7 A cone with height 14.8 cm has volume 275 cm^3 .
Calculate the radius of the cone. [

The volume, V , of a cone with radius r and height h is $V = \frac{1}{3} \pi r^2 h$.]

..... cm

(3 marks)

- 8 (a) The diagram shows a right-angled triangle ABC .



NOT TO
SCALE

The area of this triangle is 30 cm^2 .
Show that $2x^2 + 3x - 20 = 0$.

(3 marks)

(b) Use factorisation to solve the equation $2x^2 + 3x - 20 = 0$.

$x = \dots\dots\dots$ or $x = \dots\dots\dots$

(3 marks)

(c) Calculate BC .

$BC = \dots\dots\dots$ cm

(3 marks)

9 At a football match, the price of an adult ticket is $\$x$ and the price of a child ticket is $\$(x - 2.50)$.

There are 18 500 adults and 2400 children attending the football match.

The total amount paid for the tickets is \$320 040.

Find the price of an adult ticket.

\$ $\dots\dots\dots$

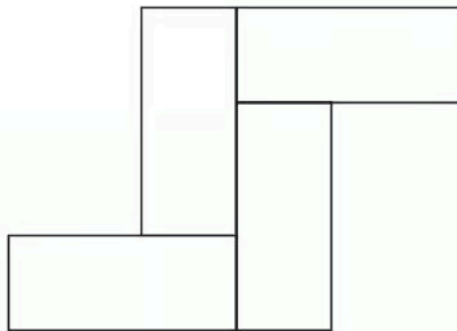
(4 marks)

10 Here is a rectangle.



The length of the rectangle is 7 cm longer than the width of the rectangle.

4 of these rectangles are used to make this 8-sided shape.

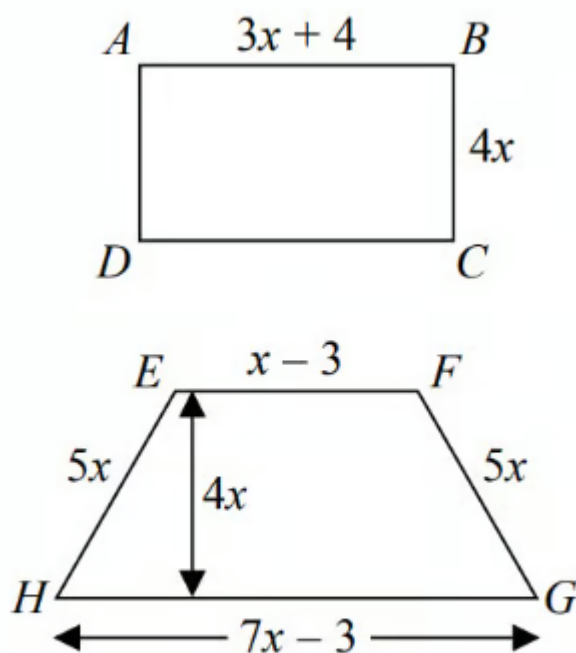


The perimeter of the 8-sided shape is 70 cm.
Work out the area of the 8-sided shape.

(5 marks)

11 $ABCD$ is a rectangle.

$EFGH$ is a trapezium.



All measurements are in centimetres.

The perimeters of these two shapes are the same.

Work out the area of the rectangle.

(5 marks)

- 12** 3 teas and 2 coffees have a total cost of £7.80
5 teas and 4 coffees have a total cost of £14.20

Work out the cost of one tea and the cost of one coffee.

(4 marks)

13 A cinema sells adult tickets and child tickets.

The total cost of 3 adult tickets and 1 child ticket is £30

The total cost of 1 adult ticket and 3 child tickets is £22

Work out the cost of an adult ticket and the cost of a child ticket.

(4 marks)

Very Hard Questions

- 1 Ahmed sells different types of cake in his shop.
The cost of each cake depends on its type and its size.

Every small cake costs \$ x and every large cake costs \$ $(2x + 1)$.

Petra spends \$20 on small coffee cakes and \$10 on large coffee cakes.
The total number of cakes is 45.

Write an equation in terms of x .

Solve this equation to find the cost of a small coffee cake.

Show all your working.

\$

(6 marks)

- 2 In a shop, the price of a monthly magazine is \$ m and the price of a weekly magazine is \$ $(m - 0.75)$.

One day, the shop receives

- \$168 from selling monthly magazines
- \$207 from selling weekly magazines.

The total number of these magazines sold during this day is 100.

- i) Show that $50m^2 - 225m + 63 = 0$.

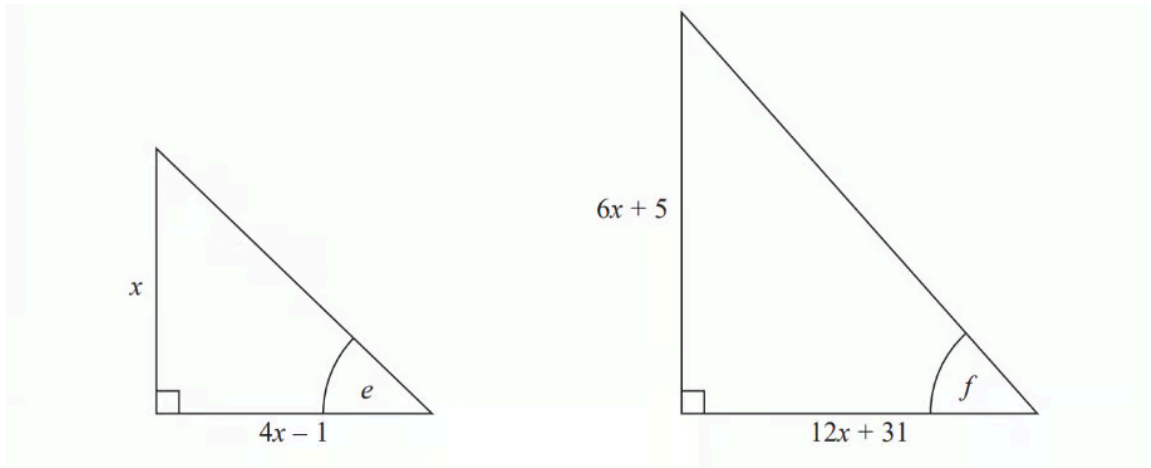
[3]

ii) Find the price of a monthly magazine.

\$ [2]

(1 mark)

3 Here are two right-angled triangles.



Given that

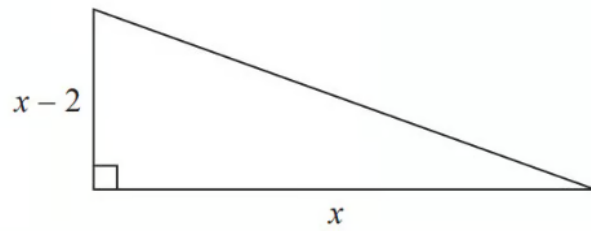
$$\tan e = \tan f$$

find the value of x .

You must show all your working.

(5 marks)

- 4 Here is a right-angled triangle.



All measurements are in centimetres.
The area of the triangle is 2.5 cm^2 .

Find the perimeter of the triangle.
Give your answer correct to 3 significant figures.
You must show all of your working.

(6 marks)

- 5 Alan invests \$200 at a rate of $r\%$ per year compound interest. After 2 years the value of his investment is \$206.46.

i) Show that $r^2 + 200r - 323 = 0$.

[3]

ii) Solve the equation $r^2 + 200r - 323 = 0$ to find the rate of interest.
Show all your working and give your answer correct to 2 decimal places.

$r = \dots\dots\dots$ [3]

(6 marks)

- 6 The total perimeter of a semicircle is 19.02 cm.
Calculate the radius of the semicircle.

..... cm

(3 marks)

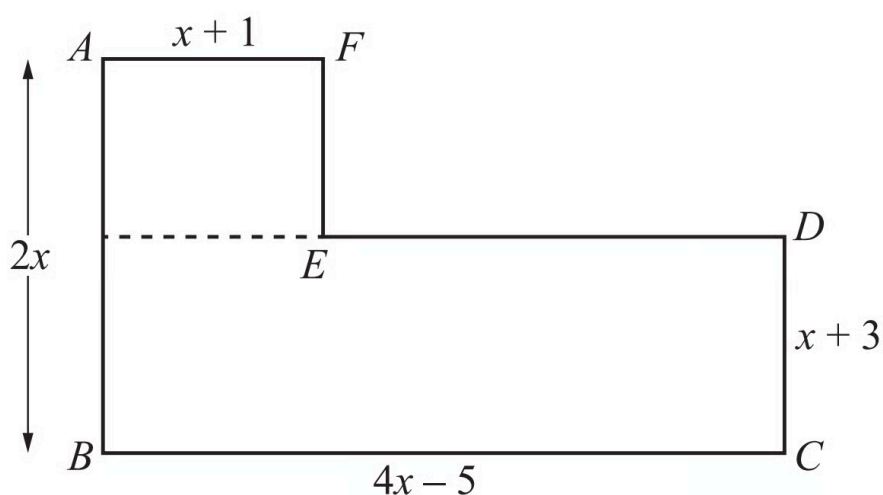
- 7 Paddy and Anna each invest \$2000 for 5 years.
Paddy earns simple interest at a rate of 1.25% per year.
Anna earns compound interest at a rate of $r\%$ per year.
At the end of 5 years, Paddy's investment is worth the same as Anna's investment.

Calculate the value of r .

$r =$

(5 marks)

8 (a) All the lengths in this question are in centimetres.



NOT TO
SCALE

The diagram shows a shape $ABCDEF$ made from two rectangles. The total area of the shape is 342 cm^2 .

Show that $x^2 + x - 72 = 0$.

(5 marks)

(b) Solve $x^2 + x - 72 = 0$ by factorisation.

$x = \dots\dots\dots$ or $x = \dots\dots\dots$

(3 marks)

(c) Work out the perimeter of the shape $ABCDEF$.

..... cm

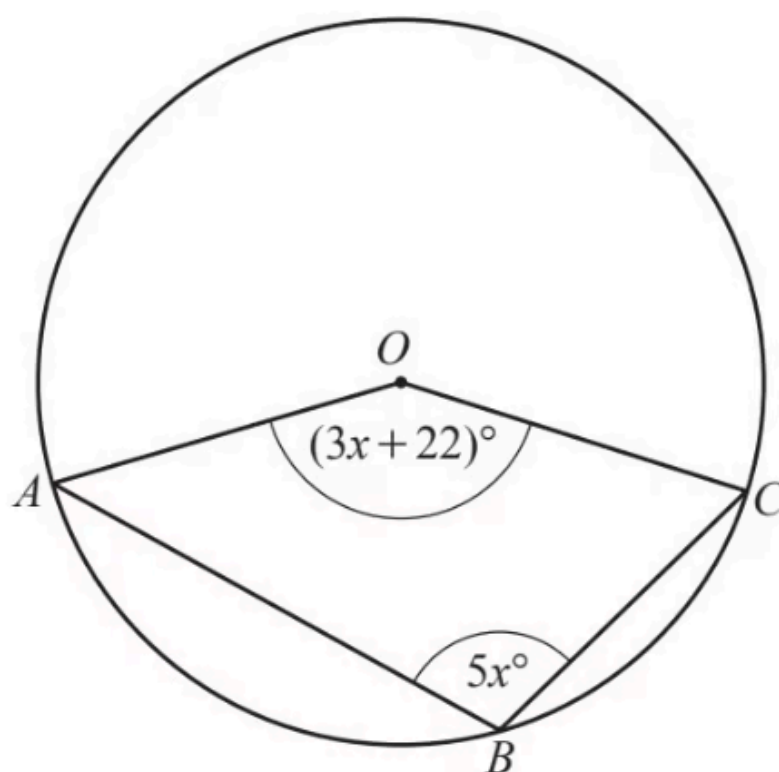
(2 marks)

(d) Calculate angle DBC .

Angle $DBC =$

(2 marks)

9



NOT TO
SCALE

A , B and C lie on the circle, centre O .
Angle $AOC = (3x + 22)^\circ$ and angle $ABC = 5x^\circ$.

Find the value of x .

$X = \dots\dots\dots$

(4 marks)

10 There are only r red counters and g green counters in a bag.

A counter is taken at random from the bag.

The probability that the counter is green is $\frac{3}{7}$ The counter is put back in the bag.

2 more red counters and 3 more green counters are put in the bag.

A counter is taken at random from the bag.

The probability that the counter is green is $\frac{6}{13}$

Find the number of red counters and the number of green counters that were in the bag originally.

(5 marks)